AP PRECALCULUS

Unit 1 • Lesson 1.8 practice

Lesson 1.8 • Rational Functions and Zeros

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{x^2 - 9}{x + 2}$?

(A) -2

(B) 3 and -3

(C) 9

(D) 3 only

Question 2

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{x^2 - 4x}{x - 4}$?

(A) 0 and 4

(B) 4 only

(C) 0 only

(D) There are none.

Question 3

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{(x+1)(x-5)}{(x-5)(x+3)}$?

(A) -1 only

(B) -1 and 5

(C) 5 only

(D) -3 and -1

Question 4

NO CALCULATOR

What is the solution set of $\dfrac{x+2}{x-1} < 0$?

(A) $(-\infty, -2)$

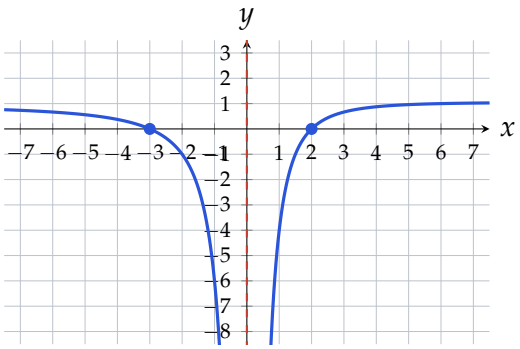
(B) $(1, \infty)$

(C) $(-\infty, -2) \cup (1, \infty)$

(D) $(-2, 1)$

Question 5 (graph) NO CALCULATOR

The graph of $r(x) = \dfrac{(x+3)(x-2)}{x^2}$ is shown. It has zeros at $x = -3$ and $x = 2$ and a vertical asymptote $x = 0$. What is the solution set of $r(x) > 0$?



(A) $(-3, 2)$
(B) $(-3, 0) \cup (0, 2)$
(C) $(-\infty, -3) \cup (2, \infty)$
(D) $(2, \infty)$

Question 6 (Free response) NO CALCULATOR

Let $r(x) = \dfrac{x^2 - x - 2}{x^2 - 9}$.

(a) Factor the numerator and denominator, and find the zeros of r.
(b) Make a sign chart for r and solve $r(x) > 0$.